2025 SSMO Relay Round 1

SMO Team

RR 1 Part 1: Let $x_1, x_2, \ldots, x_7$ be distinct integers such that the mean of $\{x_i, x_{i+1}, x_{i+2}\}$ is an integer for all integers $1 \leq i \leq 5$. Find the minimum possible positive value of $x_7 - x_1$.

RR 1 Part 2: Let $T = TNYWR$. A positive integer is called *zro* if more than half of its digits are 0. Find the sum of the first T^2 zro numbers.

RR 1 Part 3: Let $T = TNYWR$. Positive integers m and n satisfy $m^2 - n^2 = T$. What is the least possible value of $m + n$?

2025 SSMO Relay Round 2

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RR 2 Part 1: Let (p, q, r) be the real solution to the following system:

$$\begin{aligned} pq &= 20 \\ qr &= 25 \\ r + p &= 9. \end{aligned}$$

Compute $p^2 + q^2 + r^2$.

RR 2 Part 2: Let $T = TNYWR$. Let $x_1, x_2, \ldots, x_T$ be an increasing sequence of positive integers such that for every positive integer $1 \leq n \leq T$, the sum $x_1 + x_2 + \cdots + x_n$ is a multiple of n. Find the smallest possible value of $x_T - x_1$.

RR 2 Part 3: Let $T = TNYWR$. Define a *multiplicative partition* of a positive integer s as the value of a product $a_1 a_2 \cdots a_k$, where $a_1 + a_2 + \cdots + a_k = s$ and every a_i is a positive integer. Let $f(x)$ denote the maximal possible value of a multiplicative partition of x. If the sum of all possible values of $\frac{f(x+\lfloor\sqrt{T}\rfloor)}{f(x)}$ for integers $x > 1$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers, find $m + n$.

2025 SSMO Relay Round 3

SMO Team

RR 3 Part 1: Michael, Nico, and Jackson are splitting a 100 dollar bill. Michael's favorite number is 3, so he insists on paying a positive amount of dollars that is a multiple of 3. Nico and Jackson are both scared of even numbers, so they each insist on paying an odd number of dollars. In how many ways can they pay the bill?

RR 3 Part 2: Let $T = TNYWR$. In triangle ABC, angle C is a right angle. If the median from C has length T, find the maximum possible area of triangle ABC.

RR 3 Part 3: Let $T = TNYWR$. The numbers 1 and 2 are written on a board. Every second, if the numbers currently on the board are x and y, the number $x+y$ is written, and one of x or y is erased. After a finite number of seconds, the larger number on the board is $\left\lfloor \sqrt{T} \right\rfloor$. How many possible values can the smaller number take?

2025 SSMO Relay Round 4

SMO Team

RR 4 Part 1: Call a positive integer *chuzzed* if the sum of the digits in its binary representation is equal to the units digit of its base-10 representation. Similarly, call a positive integer *chopped* if its binary representation does not contain two consecutive ones. Find the number of positive integers less than 128 that are chuzzed and chopped.

RR 4 Part 2: Let $T = TNYWR$. Jonathan and Kate are playing a game with n sticks. On each turn, a player may remove 1, 2, or 3 sticks. The player who picks up the last stick loses. Kate is first to remove sticks, and both players play optimally. For how many values of n in the range $\left[T^3, 2T^3\right]$ does Kate have a winning strategy?

RR 4 Part 3: Let $T = TNYWR$. A particle moves in the coordinate plane such that at any time t, its position is

$$\left(\sum_{a=1}^{T-1} \cos(at), \sum_{a=1}^{T-1} \sin(at)\right).$$

Over the time interval $t \in (0, k]$, the particle lies on a coordinate axis at T distinct times. The minimal value of k can be expressed as $\frac{m\pi}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

2025 SSMO Relay Round 5

SMO Team

RR 5 Part 1: The numbers 17 and 25 are written on a blackboard. Every second, if the number n is currently written on the blackboard, it is replaced by $3n$ with probability p and by $2n+3$ with probability $1-p$. For example, after one second, the two numbers on the blackboard may be $3 \cdot 17 = 51$ and $2 \cdot 25 + 3 = 53$. Given that the expected value of the number descended from 25 minus the number descended from 17 after six seconds is 1000, the value of p can be expressed as $\sqrt{a} - b$, where a and b are positive integers. Find $a + b$.

RR 5 Part 2: Let $T = TNYWR$. Ethan places T red cards and T blue cards in a row so that the leftmost card is red and every second card is red. He performs a sequence of operations called *replacements*: in each replacement, he selects a card other than the leftmost one uniformly at random, discards it, and moves the leftmost card to the discarded card's position. This process is repeated until only one card remains. The probability that the final remaining card is red can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

RR 5 Part 3: Let $T = TNYWR$. How many positive integers n satisfy

$$\text{lcm}(\gcd(n, n^2 + T^2), \gcd(T, n^2 + T^2)) = \text{lcm}(n, T)?$$